# Yuehui Ma

Molecular Clouds and Interstellar Turbulence

Galactic ISM | CO Surveys | Star Formation

**Purple Mountain Observatory, CAS**
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## Professional Appointments

**Visiting Scholar** 2026.02–present
Purple Mountain Observatory, Chinese Academy of Sciences; Molecular Gas in the Milky Way Group

**Assistant Researcher** 2024.05–2026.01
Purple Mountain Observatory, Chinese Academy of Sciences; Molecular Gas in the Milky Way Group

**Assistant Researcher** 2020.02–2024.04
Purple Mountain Observatory, Chinese Academy of Sciences; Molecular Clouds and Star Formation Group

**Project Member** 2015–present
Milky Way Imaging Scroll Painting Project, Purple Mountain Observatory, Chinese Academy of Sciences

**Physics Teacher** 2010–2012
Changji No.1 High School, Xinjiang Autonomous Region

## Education

**Ph.D. in Astrophysics** 2020
**University of Chinese Academy of Sciences**
**Dissertation:** An Observational Study of the Structure and Physical Properties of Molecular Clouds via Cases and a Sample

**B.S. in Physics** 2010
**Shaanxi Normal University**

## Research Interests

- Turbulence in the interstellar medium
- Structure and dynamics of molecular clouds
- Star formation

## Grants

### As Principal Investigator (PI)

- **National Natural Science Foundation of China (NSFC), Youth Program (No. 12303033)**
  *"Study on the Dominance and Fundamental Properties of Turbulence in Galactic Molecular Clouds"*
  Jan. 2024 – Dec. 2026
- **PMO Internal Research Grant, Grants for Junior Talents (No. E2ZC201111)**
  *"Study on the Column Density Structures of Galactic Molecular Clouds"*
  Sep. 2022 – Aug. 2023

### As Participant

- **National Natural Science Foundation of China (NSFC), General Program (No. 12473026)**
  *"Star Formation from the Perspective of Complex Systems Science: A Statistical Study of Star Formation Rates in Galactic Molecular Clouds"*
  Jan. 2025 – Dec. 2028
- **NSFC General Program (No. 11973091)**

*"Multiscale Analysis of Molecular Cloud Structures and Physical Properties"*
Jan. 2020 – Dec. 2023

- **NSFC General Program (No. 11973090)**
  *"Ring Structures and Formation Mechanisms of Protoplanetary Disks"*
  Jan. 2020 – Dec. 2023
- **Jiangsu Provincial Natural Science Foundation, Youth Program (No. BK20181513)**
  *"Numerical Modeling of Chemical Differentiation in Star-Forming Regions"*
  Jul. 2022 – Jun. 2025

## Invited Talks

- **2024**, *"Martes Talk" Colloquium*, School of Astronomy and Space Science, Nanjing University, *"Turbulence Properties of Galactic Molecular Clouds Depicted by Column Density and Velocity Statistics"*, Nanjing, China.

## Honors and Awards

- **2025** Award for Research Achievements with Significant Academic Impact, Purple Mountain Observatory, CAS, based on two representative first-author papers published in 2021 and 2022
- **2023** Excellent Presentation Award for Young Scholars, Star Formation and Molecular Clouds Conference
- **2017–2019** Merit Student, University of Chinese Academy of Sciences
- **2014** Outstanding Student Leader, University of Chinese Academy of Sciences

## Skills

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|---|---|
| **Scientific Computing** | Python, R, IDL, GILDAS |
| **Astronomical Data Analysis** | Radio spectral-line surveys (CO, H I); dust extinction and multiwavelength data |
| **Turbulence Statistics** | Structure functions; power spectra; wavelet analysis; hierarchical analysis |
| **Statistical Modeling** | Probability distributions; high-order statistics; Bayesian inference; MCMC |
| **Simulation Data** | Numerical simulation data analysis; comparison with observations |
| **AI-assisted Research** | AI-agent-assisted scientific workflows for literature review, code development, data analysis, and task automation |

## Academic Visits

**Invited Academic Visit** 2023
Chalmers University of Technology, Sweden; host: Jouni Kainulainen

## Academic Service

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|---|---|
| **Journal Referee** | Astronomy & Astrophysics (A&A)<br>Publications of the Astronomical Society of Japan (PASJ) |

## Outreach

- **2014–2016** President of Student Union, PMO, CAS
- **2014–2015** Student Leader of Popular Science Association, PMO, CAS

## Refereed Journal Papers

### First/Corresponding Author Papers

[1] Suziye He, **Yuehui Ma**, Xuepeng Chen, Hongchi Wang, Miaomiao Zhang, Zhenyi Yue, and Xiangyu Ou. "Hierarchical Dynamics within Molecular Clouds: Type 4 Scaling Relations of Molecular Clouds in the Second Galactic Quadrant". In: ApJ 1010, 235 (2026), p. 235. DOI: 10.3847/1538-4357/aea8f8. arXiv: 2609.23302 [astro-ph.GA].

[2] Suziye He, **Yuehui Ma**, Hongchi Wang, Renjie Shen, Miaomiao Zhang, Chong Li, Zhenyi Yue, Xiangyu Ou, and Xuepeng Chen. "Hierarchical Structure and Self-gravity in the Rosette Molecular Cloud". In: ApJ 996.1, 62 (Jan. 2026), p. 62. DOI: 10.3847/1538-4357/ae1cc8. arXiv: 2511.02334 [astro-ph.GA].

[3] **Yuehui Ma**, Miaomiao Zhang, Hongchi Wang, Xuepeng Chen, Zhenyi Yue, Suziye He, Xiangyu Ou, and Li Sun. "Examining Turbulence in Galactic Molecular Clouds. II. Turbulence Cascade Beyond the Scale of Giant Molecular Clouds". In: ApJ 1010, 211 (2026), p. 211. DOI: 10.3847/1538-4357/aea610. arXiv: 2511.12418 [astro-ph.GA].

[4] **Yuehui Ma**, Miaomiao Zhang, Hongchi Wang, Min Fang, Zhenyi Yue, Xuepeng Chen, Ji Yang, Fujun Du, Yang Su, Suziye He, Haoran Feng, Yan Sun, Chong Li, Qing-Zeng Yan, Zhiwei Chen, Shaobo Zhang, and Xin Zhou. "Examining Turbulence in Galactic Molecular Clouds. I. A Statistical Analysis of Velocity Structures". In: ApJ 979.1, 65 (Jan. 2025), p. 65. DOI: 10.3847/1538-4357/ad9b0e. arXiv: 2501.11859 [astro-ph.GA].

[5] Renjie Shen, **Yuehui Ma**, Hongchi Wang, Suziye He, and Miaomiao Zhang. "Hierarchical Structure and Self-gravity in the Maddalena Giant Molecular Cloud". In: ApJ 971.1, 14 (Aug. 2024), p. 14. DOI: 10.3847/1538-4357/ad5347. arXiv: 2405.19766 [astro-ph.GA].

[6] **Yuehui Ma**, Hongchi Wang, Miaomiao Zhang, Chen Wang, Shaobo Zhang, Yao Liu, Chong Li, Yuqing Zheng, Lixia Yuan, and Ji Yang. "Gas Column Density Distribution of Molecular Clouds in the Third Quadrant of the Milky Way". In: ApJS 262.1, 16 (Sept. 2022), p. 16. DOI: 10.3847/1538-4365/ac7797. arXiv: 2206.03963 [astro-ph.GA].

[7] **Yuehui Ma**, Hongchi Wang, Chong Li, Lianghao Lin, Yan Sun, and Ji Yang. "Molecular Clouds in the Second Quadrant of the Milky Way Midplane from l = 104°75 to l = 119°75 and b = -5°25 to b = 5°25". In: ApJS 254.1, 3 (May 2021), p. 3. DOI: 10.3847/1538-4365/abe85c. arXiv: 2102.12199 [astro-ph.GA].

[8] **Yuehui Ma**, Hong-Chi Wang, Chong Li, and Ji Yang. "The column density structure of Orion A depicted by N-PDF". In: Research in Astronomy and Astrophysics 20.4, 060 (Apr. 2020), p. 060. DOI: 10.1088/1674-4527/20/4/60. arXiv: 1912.06984 [astro-ph.GA].

[9] **Yuehui Ma**, Hongchi Wang, Miaomiao Zhang, Chong Li, and Ji Yang. "A Large-field J = 1 - 0 Survey of CO and Its Isotopologues toward the Cassiopeia A Supernova Remnant". In: ApJ 878.1, 44 (June 2019), p. 44. DOI: 10.3847/1538-4357/ab1ea7. arXiv: 1905.10193 [astro-ph.GA].

### Co-authored Papers

[10] Xuepeng Chen, Li Sun, **Yuehui Ma**, Min Fang, Fujun Du, Ji Yang, Yang Su, Yan Sun, Xin Zhou, Qing-Zeng Yan, Shaobo Zhang, Lixia Yuan, Shiyu Zhang, and Xin Liu. "Discovery of A Giant Spiral Molecular Cloud in the Galactic Anticenter. II. A Black Hole and Jet System". In: AJ 172.2, 101 (Aug. 2026), p. 101. DOI: 10.3847/1538-3881/ae83a0.

[11] Xuepeng Chen, Li Sun, **Yuehui Ma**, Min Fang, Fujun Du, Ji Yang, Yang Su, Yan Sun, Xin Zhou, Qing-Zeng Yan, Shaobo Zhang, Lixia Yuan, Shiyu Zhang, Xin Liu, and Mwisp Group. "Discovery of a Giant Spiral Molecular Cloud in the Galactic Anticenter". In: AJ 172.1, 4 (July 2026), p. 4. DOI: 10.3847/1538-3881/ae6800.

[12] Ruilin Xia, Yang Su, Shiyu Zhang, Xuepeng Chen, Ji Yang, Yan Gong, **Yuehui Ma**, Yan Sun, Min Fang, Fujun Du, Shaobo Zhang, Xin Zhou, Lixia Yuan, Qing-Zeng Yan, Li Sun, and Jiancheng Feng. "CO Structures with Narrow Lines in Nearby Quiescent Regions". In: ApJL 1007.1, L21 (Aug. 2026), p. L21. DOI: 10.3847/2041-8213/ae9091. arXiv: 2608.07321 [astro-ph.GA].

[13] Ji Yang, Qing-Zeng Yan, Yang Su, Shaobo Zhang, Xin Zhou, Yan Sun, Yiping Ao, Xuepeng Chen, Zhiwei Chen, Fujun Du, Min Fang, Yan Gong, Zhibo Jiang, Shengyu Jin, Binggang Ju, Chong Li, Yingjie Li, Yi Liu, Dengrong Lu, Chunsheng Luo, **Yuehui Ma**, Ruiqing Mao, Jixian Sun, Chen Wang, Hongchi Wang, Min Wang, Min Wang (Qinghai), Xindong Wang, Wenting Xu, Ye Xu, Kun Yan, Ping Yan, Lixia Yuan, Miaomiao Zhang, and Yongxing Zhang. "The Milky Way Imaging Scroll Painting Survey: Data Release 1". In: ApJS 282.2, 65 (Feb. 2026), p. 65. DOI: 10.3847/1538-4365/ae29e7. arXiv: 2512.08260 [astro-ph.GA].

[14] Yang Su, Shiyu Zhang, Yan Sun, Ji Yang, Fujun Du, Min Fang, Qing-Zeng Yan, Shaobo Zhang, Zhiwei Chen, Xuepeng Chen, Xin Zhou, Lixia Yuan, and **Yuehui Ma**. "Gas Transfer between the Inner 3 kpc Disk and the Galactic Central Molecular Zone". In: ApJ 984.2, 109 (May 2025), p. 109. DOI: 10.3847/1538-4357/adc38e. arXiv: 2503.12313 [astro-ph.GA].

[15] Haoran Feng, Zhiwei Chen, Zhibo Jiang, **Yuehui Ma**, Yang Yang, Shuling Yu, Dongqing Ge, Wei Zhou, Fujun Du, Chen Wang, Shiyu Zhang, Yang Su, and Ji Yang. "Revisiting the Velocity Dispersion–Size Relation in Molecular Cloud Structures". In: Research in Astronomy and Astrophysics 24.11, 115018 (Nov. 2024), p. 115018. DOI: 10.1088/1674-4527/ad89a9. arXiv: 2410.15034 [astro-ph.GA].

[16] Yan Sun, Ji Yang, Qing-Zeng Yan, Shaobo Zhang, Yang Su, Xuepeng Chen, Xin Zhou, **Yuehui Ma**, and Lixia Yuan. "Molecular Clouds in the Outer Milky Way Disk: Sample, Integrated Properties, and Radial Trends with Galactocentric Radius". In: ApJS 275.2, 35 (Dec. 2024), p. 35. DOI: 10.3847/1538-4365/ad8237.

[17] Shiyu Zhang, Yang Su, Xuepeng Chen, Min Fang, Qing-Zeng Yan, Shaobo Zhang, Yan Sun, Xiaolong Wang, Haoran Feng, **Yuehui Ma**, Miaomiao Zhang, Zi Zhuang, Xin Zhou, Zhiwei Chen, and Ji Yang. "The Multilayer Nature of Molecular Gas toward the Cygnus Region". In: AJ 167.5, 220 (May 2024), p. 220. DOI: 10.3847/1538-3881/ad2fcb. arXiv: 2403.00061 [astro-ph.GA].

[18] Chong Li, Keping Qiu, Di Li, Hongchi Wang, Yue Cao, Junhao Liu, **Yuehui Ma**, and Chenglin Yang. "H I Self-absorption toward the Cygnus X North: From Atomic Filament to Molecular Filament". In: ApJL 948.2, L17 (May 2023), p. L17. DOI: 10.3847/2041-8213/accf99. arXiv: 2305.10795 [astro-ph.GA].

[19] Chunxue Li, Hongchi Wang, **Yuehui Ma**, Lianghao Lin, Yang Su, Chong Li, Yan Sun, Xin Zhou, and Ji Yang. "Molecular Clouds in the Galactic Plane from l = [59.°75, 74.°75] and b = [-5.°25, +5.°25]". In: ApJS 267.2, 30 (Aug. 2023), p. 30. DOI: 10.3847/1538-4365/acd9a7. arXiv: 2305.18043 [astro-ph.GA].

[20] Chen Wang, Haoran Feng, Ji Yang, Xuepeng Chen, Yang Su, Qing-Zeng Yan, Fujun Du, **Yuehui Ma**, and Jiajun Cai. "The Molecular Clouds in a Section of the Third Galactic Quadrant: Observational Properties and Chemical Abundance Ratio between CO and its Isotopologues". In: AJ 166.3, 121 (Sept. 2023), p. 121. DOI: 10.3847/1538-3881/acebdd. arXiv: 2308.10726 [astro-ph.GA].

[21] Chen Wang, Haoran Feng, Ji Yang, Xuepeng Chen, Yang Su, Qinq-Zeng Yan, Fujun Du, **Yuehui Ma**, and Jiajun Cai. "The Molecular Clouds in a Section of the Third Galactic Quadrant: Cloud Catalog". In: AJ 165.3, 106 (Mar. 2023), p. 106. DOI: 10.3847/1538-3881/acafee.

[22] Lixia Yuan, Ji Yang, Fujun Du, Xunchuan Liu, Yang Su, Qing-Zeng Yan, Xuepeng Chen, Yan Sun, Shaobo Zhang, Xin Zhou, and **Yuehui Ma**. "On the Spatial Distribution of $^{13}$CO Structures within $^{12}$CO Molecular Clouds". In: ApJ 944.1, 91 (Feb. 2023), p. 91. DOI: 10.3847/1538-4357/acac26. arXiv: 2212.02066 [astro-ph.GA].

[23] Xin Zhou, Yang Su, Ji Yang, Xuepeng Chen, Yan Sun, Zhibo Jiang, Min Wang, Hongchi Wang, Shaobo Zhang, Ye Xu, Qingzeng Yan, Lixia Yuan, Zhiwei Chen, Yiping Ao, and **Yuehui Ma**. "A Systematic Study of Associations between Supernova Remnants and Molecular Clouds". In: ApJS 268.2, 61 (Oct. 2023), p. 61. DOI: 10.3847/1538-4365/acee7f. arXiv: 2308.03484 [astro-ph.GA].

[24] Chun-Xue Li, Hong-Chi Wang, **Yuehui Ma**, Miao-Miao Zhang, Chong Li, and Yu-Qing Zheng. "Molecular Clouds Associated with H II Regions and Candidates within l = 106.°65 to 109.°50 and b = -1.°85 to 0.°95". In: Research in Astronomy and Astrophysics 22.4, 045008 (Apr. 2022), p. 045008. DOI: 10.1088/1674-4527/ac52a0. arXiv: 2202.04264 [astro-ph.GA].

[25] Jia Sun, Robert A. Gutermuth, Hongchi Wang, Miaomiao Zhang, Shuinai Zhang, **Yuehui Ma**, Xinyu Du, and Min Long. "Deep near-infrared survey towards the W40 and Serpens South region in the Aquila Rift: A comprehensive catalogue of young stellar objects". In: MNRAS 516.4 (Nov. 2022), pp. 5244–5257. DOI: 10.1093/mnras/stac2191. arXiv: 2207.09041 [astro-ph.SR].

[26] Lixia Yuan, Ji Yang, Fujun Du, Yang Su, Xunchuan Liu, Shaobo Zhang, Yan Sun, Xin Zhou, Qing-Zeng Yan, and **Yuehui Ma**. "Molecular Gas Structures Traced by $^{13}$CO Emission in the 18,190 $^{12}$CO Molecular Clouds from the MWISP Survey". In: ApJS 261.2, 37 (Aug. 2022), p. 37. DOI: 10.3847/1538-4365/ac739f. arXiv: 2205.14858 [astro-ph.GA].

[27] Lixia Yuan, Ji Yang, Fujun Du, Xunchuan Liu, Shaobo Zhang, Zehao Lin, Jingfei Sun, Qing-Zeng Yan, **Yuehui Ma**, Yang Su, Yan Sun, and Xin Zhou. "A Morphological Classification of 18,190 Molecular Clouds Identified in $^{12}$CO Data from the MWISP Survey". In: ApJS 257.2, 51 (Dec. 2021), p. 51. DOI: 10.3847/1538-4365/ac242a. arXiv: 2108.12110 [astro-ph.GA].

[28] Yu-Qing Zheng, Hong-Chi Wang, **Yuehui Ma**, and Chong Li. "Symmetry properties and widths of the filamentary structures in the Orion A giant molecular cloud". In: Research in Astronomy and Astrophysics 21.8, 188 (Oct. 2021), p. 188. DOI: 10.1088/1674-4527/21/8/188. arXiv: 2103.10637 [astro-ph.GA].

[29] Chakali Eswaraiah, Di Li, Manash R. Samal, Jia-Wei Wang, **Yuehui Ma**, Shih-Ping Lai, Annie Zavagno, Tao-Chung Ching, Tie Liu, Kate Pattle, Derek Ward-Thompson, Anil K. Pandey, and Devendra K. Ojha. "Unveiling the Importance of Magnetic Fields in the Evolution of Dense Clumps Formed at the Waist of Bipolar H II Regions: A Case Study of Sh 2-201 with JCMT SCUBA-2/POL-2". In: ApJ 897.1, 90 (July 2020), p. 90. DOI: 10.3847/1538-4357/ab83f2. arXiv: 1912.10188 [astro-ph.GA].

[30] Chong Li, Hong-Chi Wang, Yuan-Wei Wu, **Yuehui Ma**, and Liang-Hao Lin. "Assessing the performance of molecular gas clump identification algorithms". In: Research in Astronomy and Astrophysics 20.3, 031 (Mar. 2020), p. 031. DOI: 10.1088/1674-4527/20/3/31. arXiv: 1910.07692 [astro-ph.GA].

[31] Chong Li, Hongchi Wang, Miaomiao Zhang, **Yuehui Ma**, and Lianghao Lin. "The Molecular Clouds Associated with the H II Regions/Candidates between L = 207°7 and L = 211°7". In: ApJS 249.2, 27 (Aug. 2020), p. 27. DOI: 10.3847/1538-4365/ab9cc0.

[32] Chakali Eswaraiah, Shih-Ping Lai, **Yuehui Ma**, Anil K. Pandey, Jessy Jose, Zhiwei Chen, Manash R. Samal, Jia-Wei Wang, Saurabh Sharma, and D. K. Ojha. "Polarimetric and Photometric Investigation of the Dark Globule LDN 1225: Distance, Extinction Law, and Magnetic Fields". In: ApJ 875.1, 64 (Apr. 2019), p. 64. DOI: 10.3847/1538-4357/ab0a0c. arXiv: 1812.08920 [astro-ph.GA].

[33] Yingjie Li, Ye Xu, Yan Sun, Qing-Zeng Yan, **Yuehui Ma**, and Ji Yang. "CO Outflow Candidates toward the W3/4/5 Complex I: The Sample and Its Spatial Distribution". In: ApJS 242.2, 19 (June 2019), p. 19. DOI: 10.3847/1538-4365/ab1e55. arXiv: 1905.02018 [astro-ph.GA].

[34] Chen Wang, Ji Yang, Yang Su, Fujun Du, **Yuehui Ma**, and Shaobo Zhang. "Molecular Gas toward the Gemini OB1 Molecular Cloud Complex. III. Chemical Abundance". In: ApJS 243.2, 25 (Aug. 2019), p. 25. DOI: 10.3847/1538-4365/ab2d2e.

[35] Chong Li, Hongchi Wang, Miaomiao Zhang, **Yuehui Ma**, Min Fang, and Ji Yang. "A Large-scale Survey of CO and Its Isotopologues toward the Rosette Molecular Cloud". In: ApJS 238.1, 10 (Sept. 2018), p. 10. DOI: 10.3847/1538-4365/aad963. arXiv: 1808.09189 [astro-ph.GA].